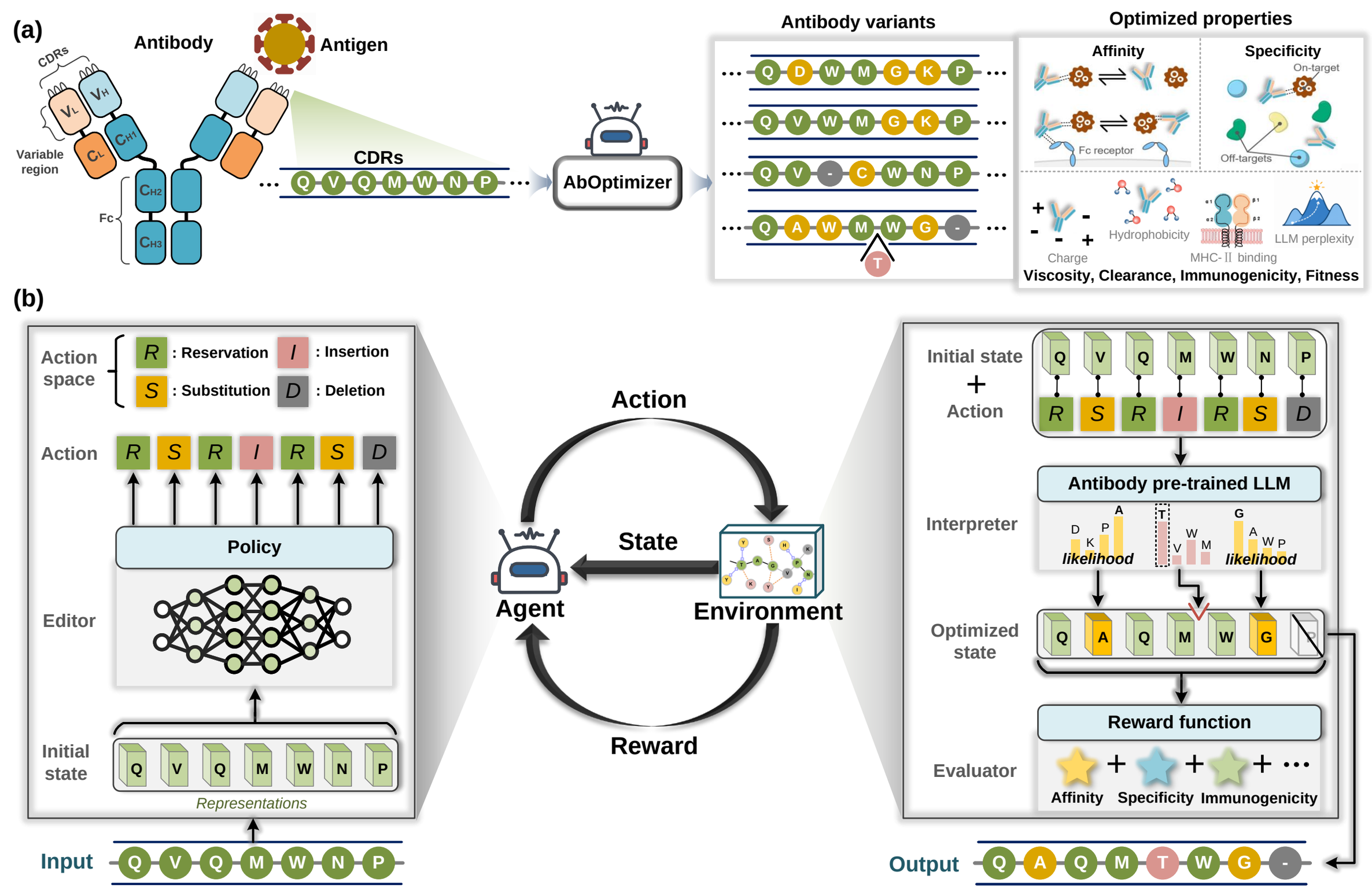

(a)
Antibody
Antigen
CDRs
Variable region
Fc
CDRs
Q V Q M W N P
AbOptimizer
Antibody variants
Q D W M G K P
Q V W M G K P
Q V - C W N P
Q A W M W G -
T
Optimized properties
Affinity
Fc receptor
Specificity
On-target
Off-targets
Charge
Hydrophobicity
MHC-Ⅱ binding
LLM perplexity
Viscosity, Clearance, Immunogenicity, Fitness
(b)
Action space
R : Reservation
I : Insertion
S : Substitution
D : Deletion
Action
R S R I R S D
Policy
Editor
Initial state
Q V Q M W N P
Representations
Input
Q V Q M W N P
Action
State
Agent
Environment
Reward
Initial state
+
Action
Q V Q M W N P
R S R I R S D
Antibody pre-trained LLM
Interpreter
likelihood
likelihood
Optimized state
Q A Q M W G
Reward function
Evaluator
Affinity
Specificity
Immunogenicity
Output
Q A Q M T W G -